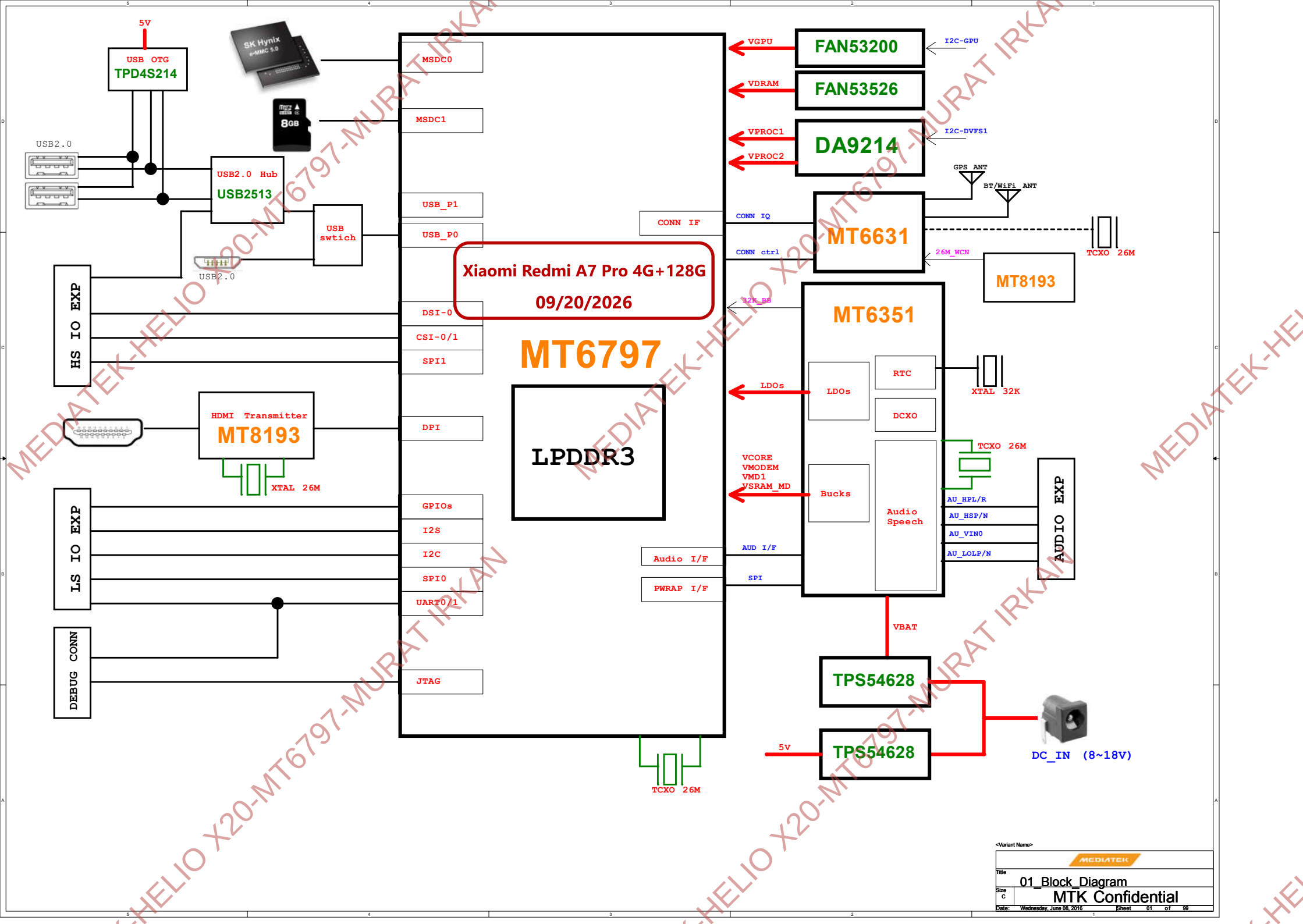




Date	Category	Item
2016.01.13	General	Preliminary release
2016.01.15	General	1. [Power] Change PL901 and PL902 to 3.3uH / SPM4015. 2. [ME] Add FD900,FD901,FD902,FD903,H901,H902,H903,H904.
2016.01.29	General	1. [DSI] Change DSI-0 to HS IO expansion and remove DSI-1. 2. [CSI] Change CSI-2 to CSI-1. 3. [GPIO] Move GPIOs on DPI interface to BPI_BUS. 4. [HDMI] Change ADV7533 to SiI9024A. 5. Modify block diagram. 6. [Power] Remove U1002~U1004, U916,U918,U919, U2302. 7. [CLK] Remove U3202. 8. [Audio] Remove U6101. 9. [Memory] Change eMMC to 8GB, DDR to 16Gb. 10. [Connector] Change CON70001 to 55510-140LF. 11. [Charging] Remove pulse charger and battery connector. 12. [RTC] Remove GB2001. 13. [Debug] Change CON9001 to NC.
2016.02.18	General	1. [HDMI] Change HDMI to MT8193.
2016.02.26	General	1. [Passive] Remove CE901,D6402. 2. [USB] Modify symbols of U6402, U6403. 3. [Connector] Change J901 to LM-DC-D049;Remove J903,J904.
2016.03.01	General	1. [Connsys] Change U5002 from ACPF-7124 to DEA162450BT-1288B2.
2016.03.14	General	1. [HDMI] Update U6502 MT8193 PCB Footprint;Updata Q6501,Q6502 Symbol;ADD TP6505,TP6506 2. [Connsys] NC R5061,Add R5001, R5002, U5001.
2016.03.15	General	1. [Connsys] ADD CON5005,CON5006,R5071,R5072,R5073,R5074.
2016.03.23	General	1. [Connsys] Change ANT5001 To chip ANT ANT016008LCD2442MA1, Change CON5005,CON5006 to MM5829-2700RJ4.
2016.05.31	General	1. [KEY_LED_EXT26M] Add SW3204,SW3205,D3201,D3202,R3225,R3226. 2. [Connsys] Update U5001 SCH symbol. 3. [SD Card] Swap pin 1 and 2 of ESD4104,ESD4105,ESD4106,ESD4107,ESD4108,ESD4109.
2016.06.07	General	1. [USB] Add switch between USB device and USB hub 2. [Camera] Change CSI-1 from 2*2-lane mode to 4-lane mode

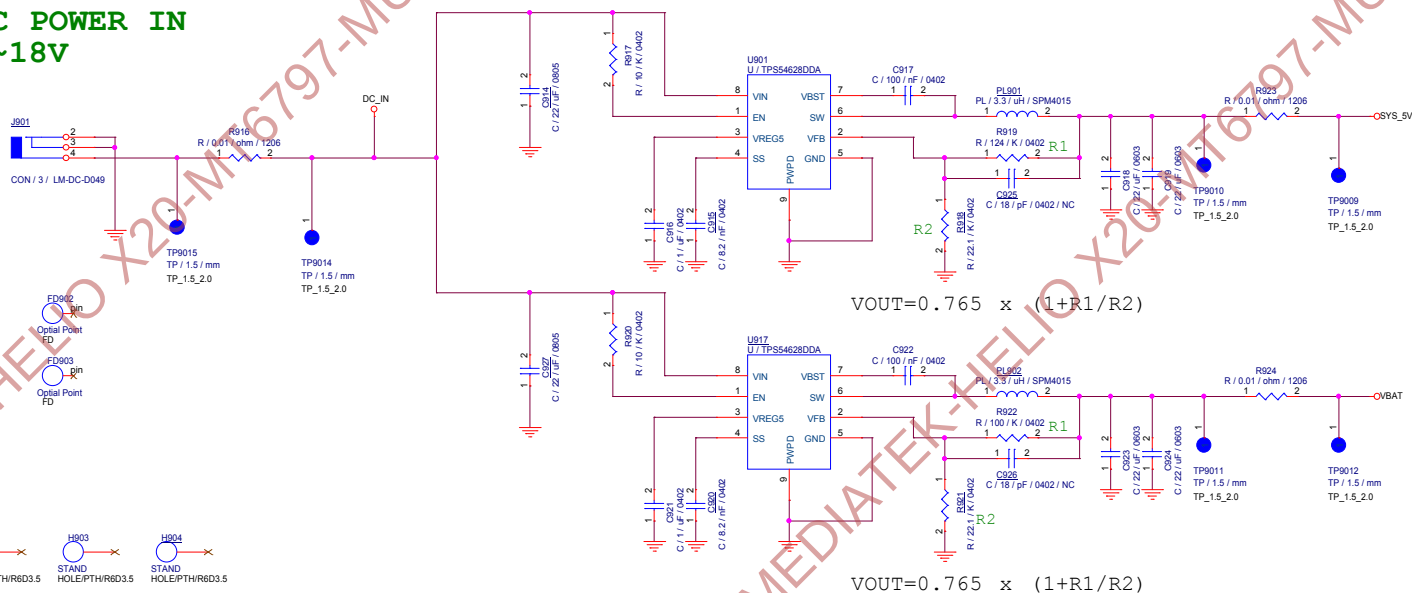
Variant Name	
MEDIATEK	
Title	05_Change_Notice
Size	MTK Confidential
Date: Wednesday, June 08, 2016	Sheet 05 of 89

Common EVB features

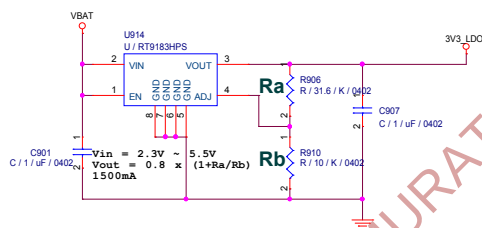
DC POWER IN 8~18V

FD

HOLE



3V3



MediaTek Helio X20 MT6797

09/20/2026

<Variant Name>

MEDIATEK	
Title	09_Common LDO Power
Size	MTK Confidential
Date	Wednesday, June 08, 2016
Sheet	09 of 09

U1001E

MT6797-MWFPPOP-950P

VCCK

PROC 2

PROC 1

GPU

MD

SRAMs

D

V

C

B

A

GND

D

C

B

A

GND

D

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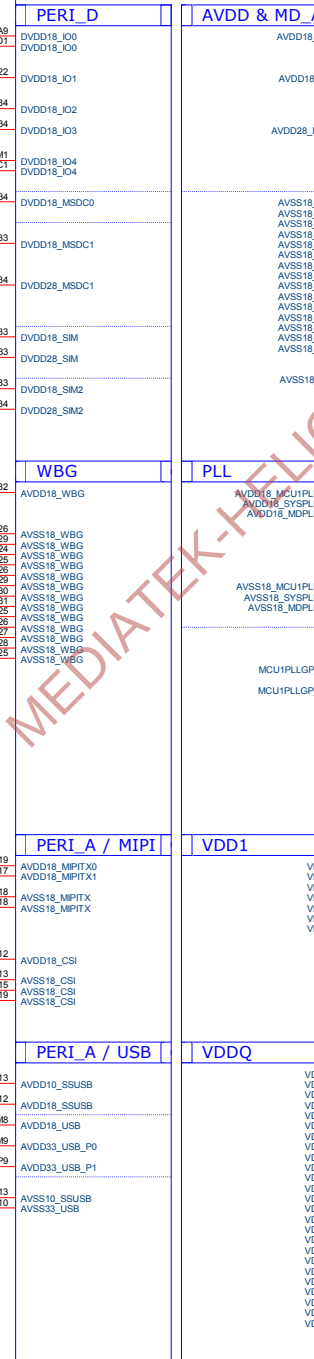
C

B

MT6797-MWFPPOP-950P



MT6797-MWFPPOP-950P



Schematic design notice of "11_BB_POWER" page.

Note 11-1: Reserve 1uF capacitor in VCC18IO0 for MHL.

Note 11-2: AVDD28_DAC (F1 ball) must be powered by "VTCXO28_PMU".

Note 11-3: The placement of power/de-coupling capacitors of DVDD18_MSDC0/1 & DVDD28_MSDC1 should be placed as close to its power ball as possible within 150mil.

Note 11-4: Analog GND that should be connected to main GND directly.

Note 11-5: Differential pair of VDRAM_PMU remote sense should be placed in the farthest power plane from MT6351 point of view.

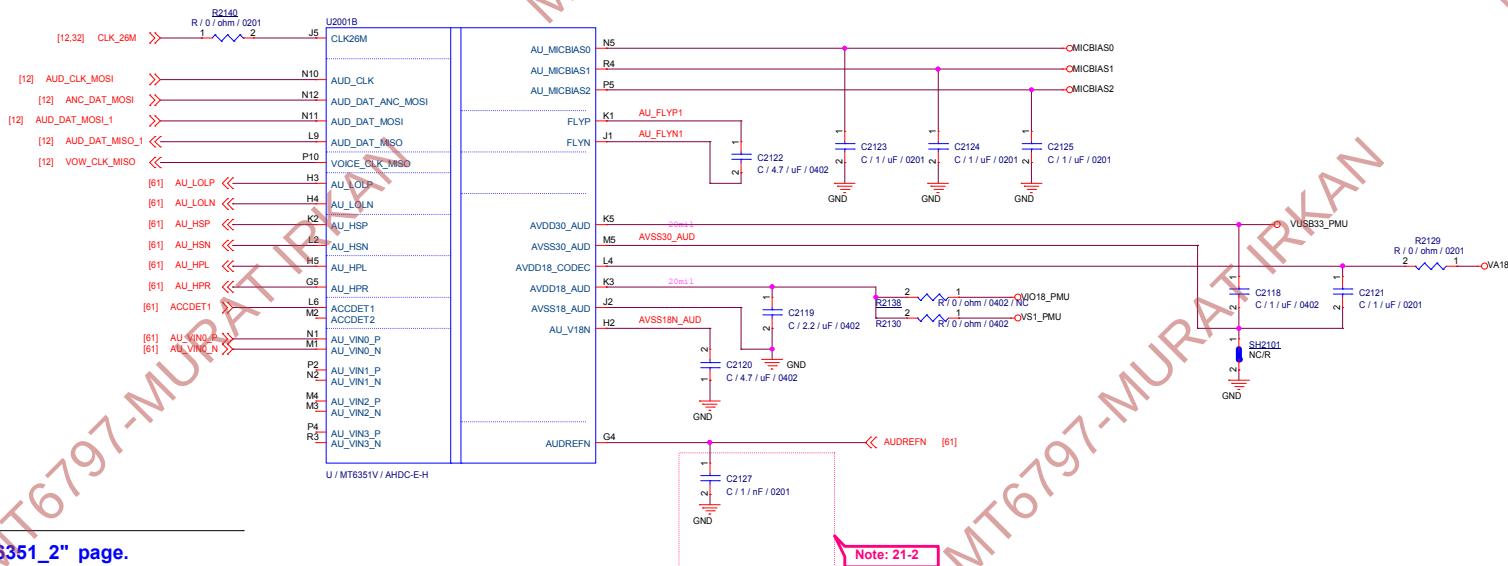
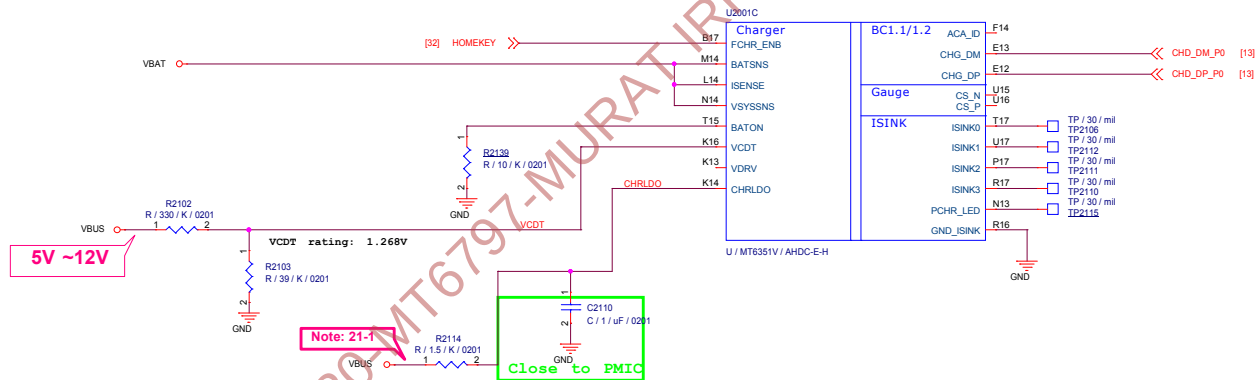
Note 11-6: Connect AVDD33_USB_P0 to "VUSB33_PMU" for USB application; Connect AVDD33_USB_P1 to "VSIM1_PMU" for IC-USB / Samrt card application.

Note 11-7: BOM option to select MT6351's VGPU or 3rd party PMIC as VDRAM (1.2V) power

MT6351's VGPU as VDRAM	C1119	C1120
	47uF	22uF
3rd party PMIC as VDRAM	NC / DNI	NC / DNI

Note: NC / DNI = No connect / Do not install.

<Variant Name>



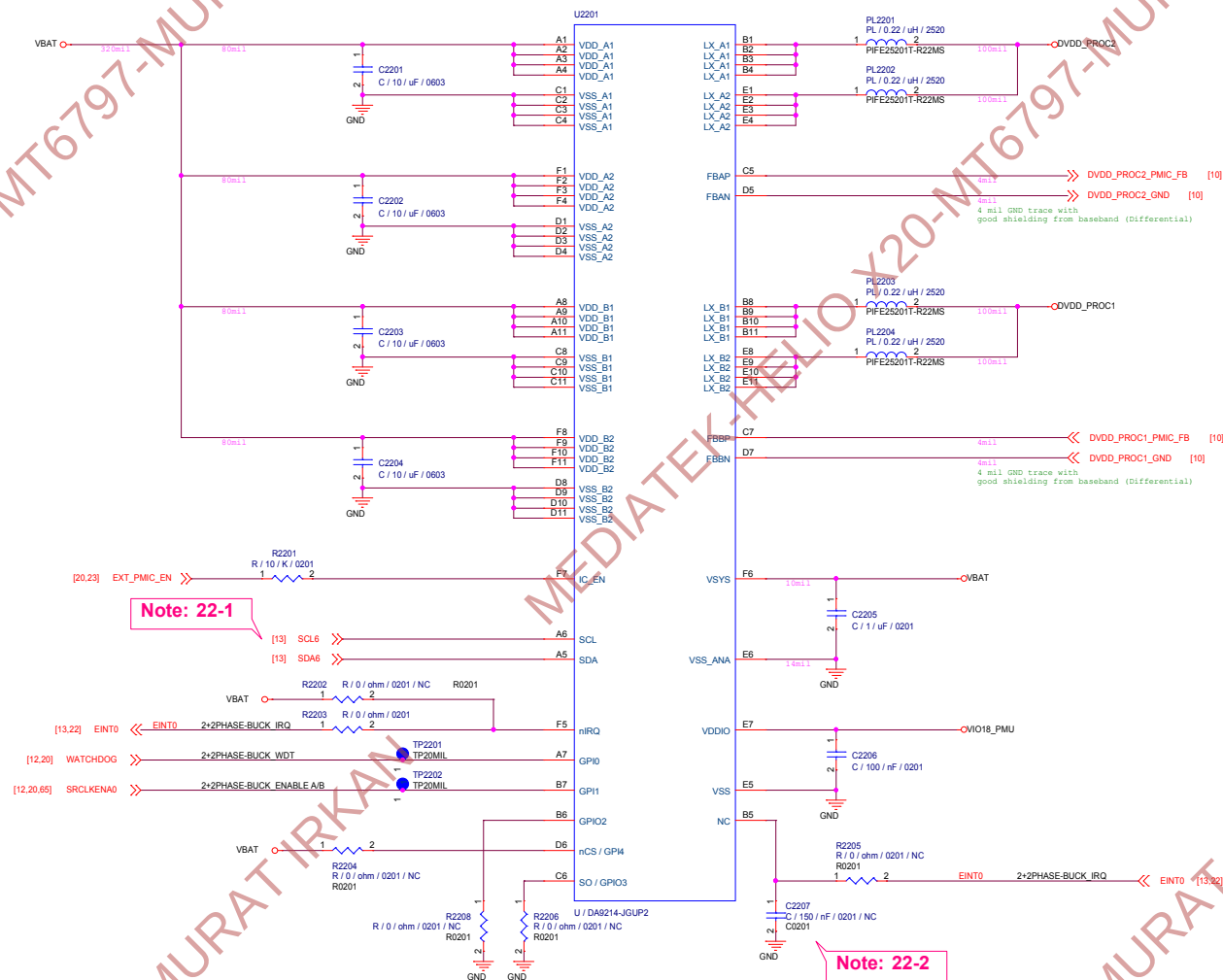
Schematic design notice of "21_POWER_MT6351_2" page.

Note 21-1: Reserve 1.5K in order to give additional power to turn on charger LED driver while low battery.

VPROC Buck

MT6313 / 2+2Phase Buck I2C address: 0X6B (Write:0xD6, Read:0xD7)

DA9214 / 2+2Phase Buck I2C address: 0X68 (Write:0xD0, Read:0xD1)



Schematic design notice of "22_POWER_VPROCS" page.

Note 22-1: Buck EN is controlled by SRCLKEN0 or I2C

Note 22-2: BOM option of 2+2 phase buck

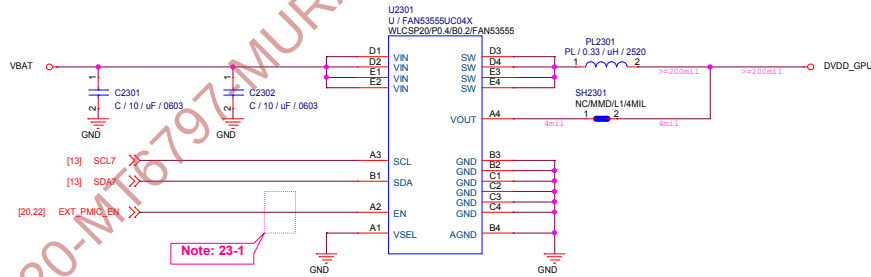
	R2202	R2203	R2204	R2205	R2206	R2208	C2207
DA9214	NC / DNI	0-ohm	NC / DNI	NC / DNI	NC / DNI	NC / DNI	150nF / NC
2nd source	0-ohm	NC / DNI	0-ohm	0-ohm	0-ohm	0-ohm	NC / DNI
NC / DNI = No connect / Do not install.							

<Variant Name>

MEDIATEK	
Title	22_POWER_VPROCS
Size	MTK Confidential
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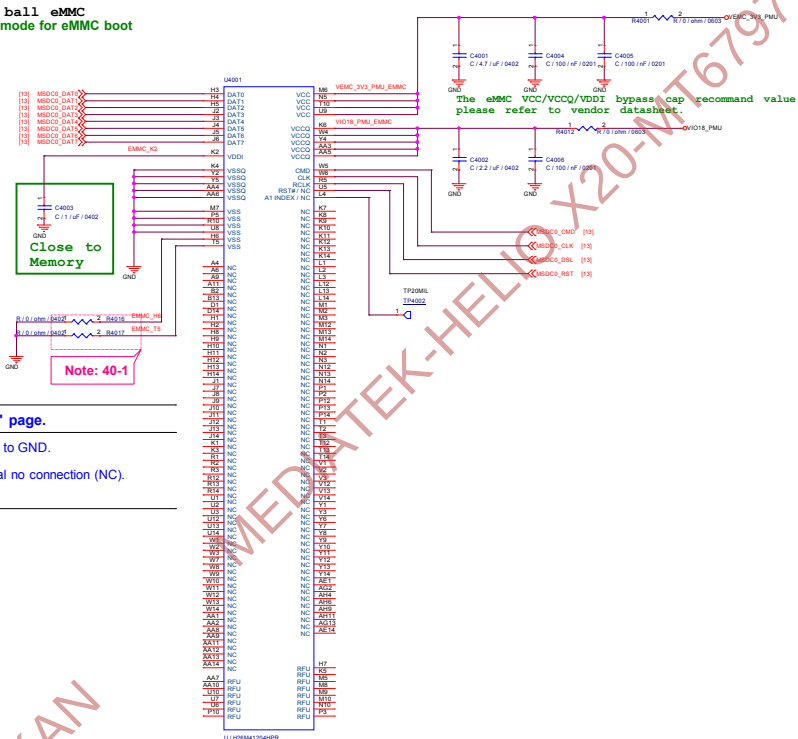
FAN53200 / Buck I2C address: 0X60 (Write:0xC0, Read:0xC1)

FAN53200 / Buck I2C address: 0X60 (Write:0xC0, Read:0xC1)



Schematic design notice of "23_POWER_VGPU_VM" page.

Note 23-1: FAN53200's EN pin is driven by MT6351.

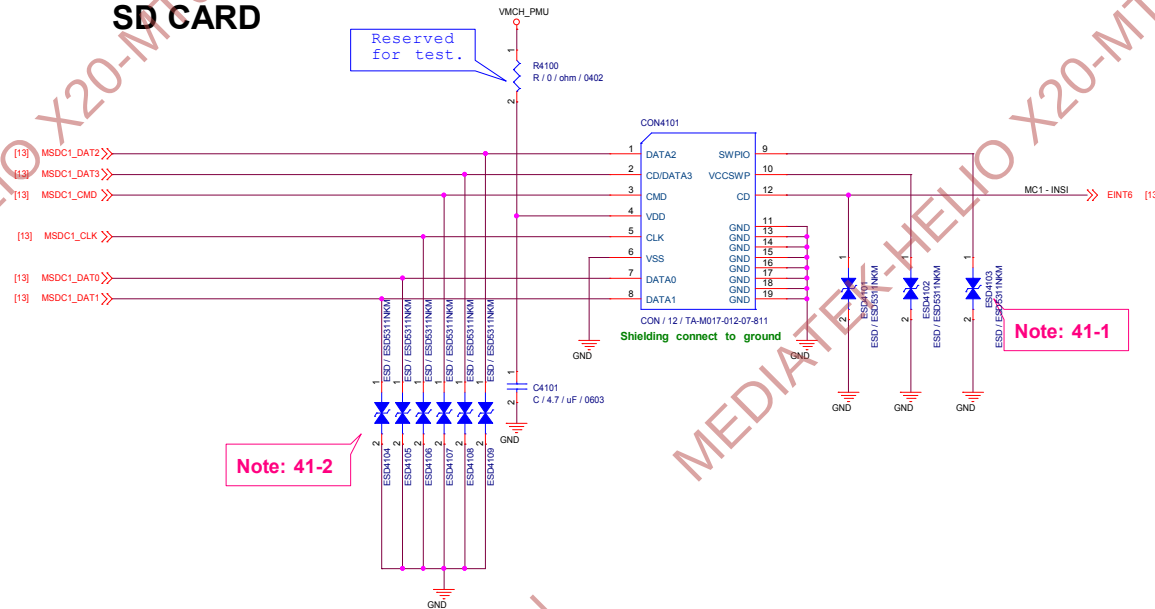


Note 40-1: For eMMC 5.0, connect eMMC's H6 & T5 pin to GND.

Note 40-1: For eMMC 5.0, connect eMMC's H6 & T5 pin to GND.

For eMMC 4.5, check eMMC's H6 & T5 is real no connection (NC).

SD CARD



Schematic design notice of "41_MEMORY_SD Card" page.

Note 41-1: The equivalent capacitance of ESD protection device must be $\leq 1\text{pF}$
 -- otherwise it will result in NFC card mode function fail.

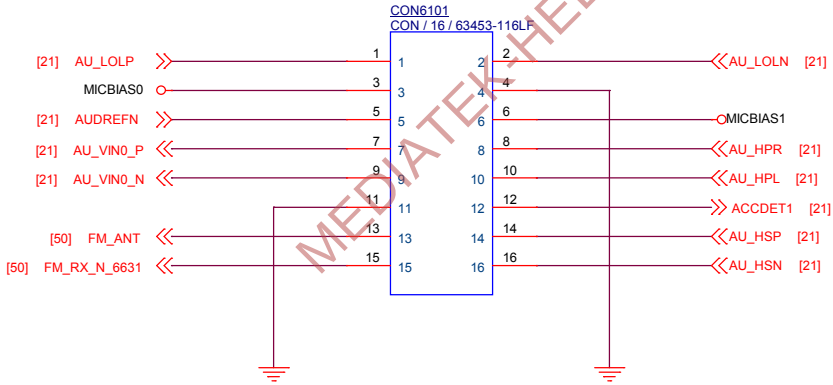
Note 41-2: Depends on system design to add ESD protection component or not.

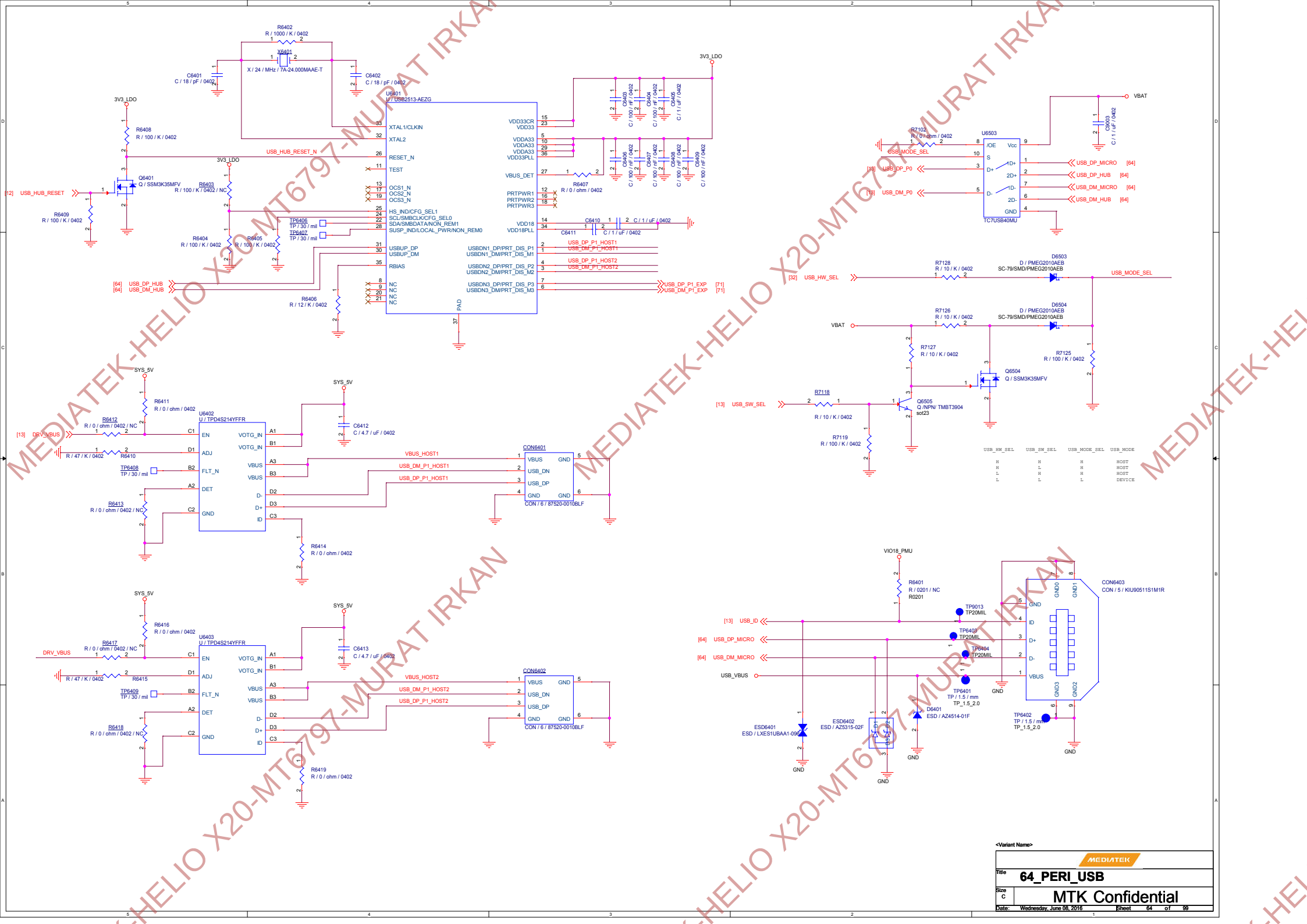
<Variant Name>

MEDIATEK	
Title	41_MEMORY_SD Card
Size	MTK Confidential
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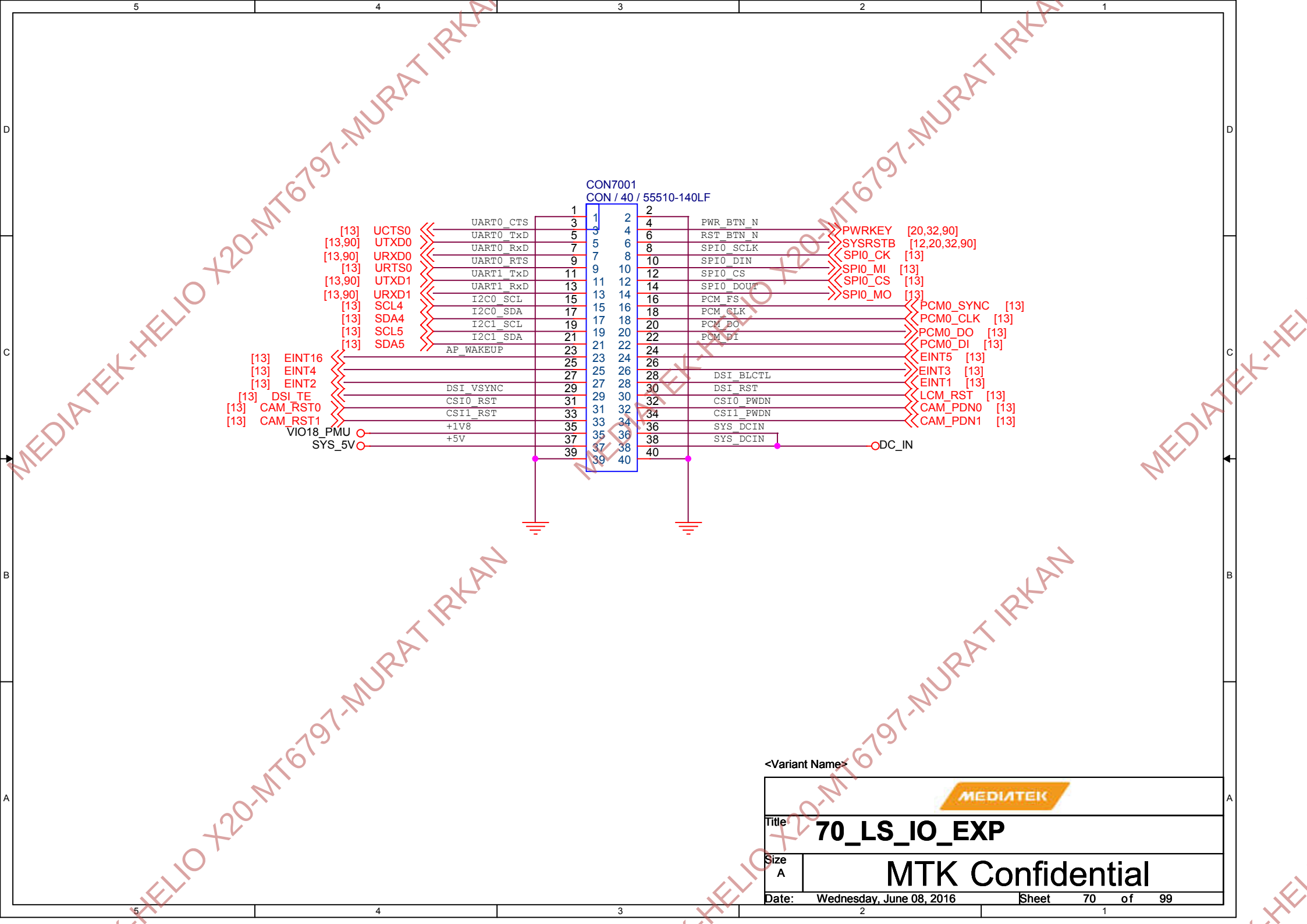


Audio Expansion







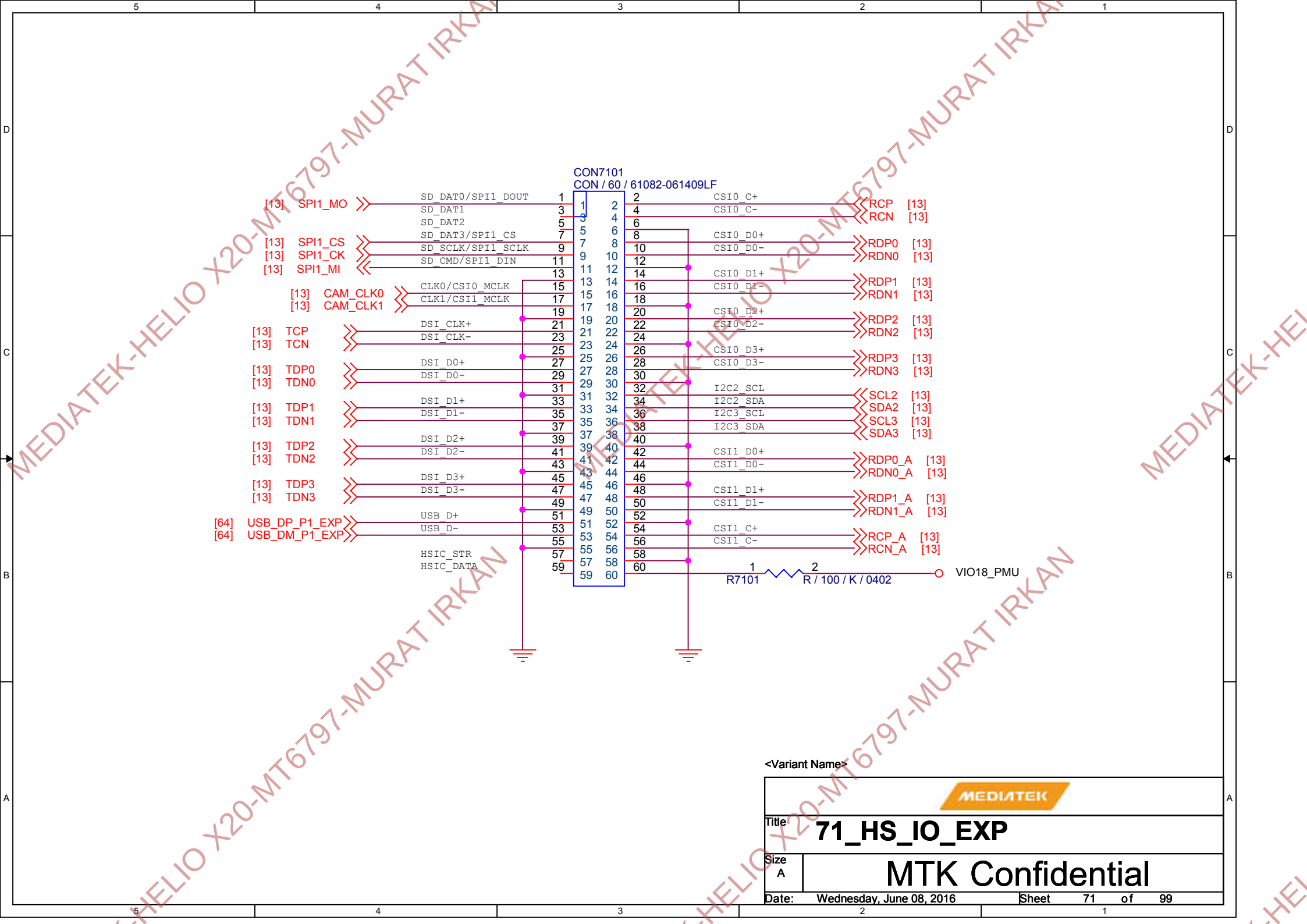


<Variant Name>



Title 70_LS_IO_EXP

Size A MTK Confidential



<Variant Name>

MEDIATEK

Title 71_HS_IO_EXP

Size A MTK Confidential

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MTK Debug I/O

